

Hack for Green **भारत**

Team Name - Xenovate
presenting

PULSE PREDICT

Predicting Failures Before They Happen

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Real time monitoring reduces unexpected downtime and costly failures

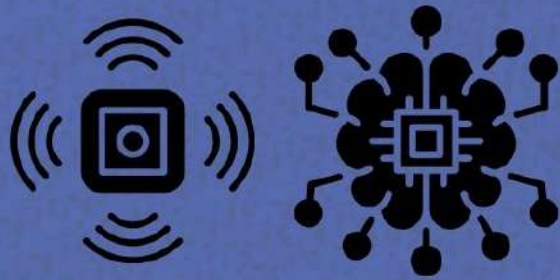




OCCURENCE OF PROBLEM

- Unexpected **equipment overheating** and **vibration spikes** lead to sudden motor or belt failures without early warning.
- Lack of continuous real time monitoring prevents **detection of gradual wear, misalignment, and internal component damage.**
- Reactive maintenance approach causes production downtime, **financial losses**, and increased safety risks for workers.

OUR SOLUTION -



Technologies Used

Integrating Sensors & Artificial Intelligence

- *PulsePredict monitors machine health in real time and predicts failures early.*
- *It collects temperature, vibration, and pressure data from equipment continuously.*
- *Streaming analytics detect overheating, abnormal vibration, and wear patterns.*
- *The system alerts technicians before breakdowns cause production downtime.*
- *An AI assistant explains possible causes and recommends maintenance actions.*

- **Kafka / MQTT** : To simulate and transmit continuous sensor data streams.
- **Pathway Streaming Engine** : To process live data and detect anomalies in real time.
- **Feature Engineering** : Rolling averages, spike detection, and trend analysis for smarter predictions.
- **Machine Learning Model** : To estimate failure risk based on sensor behaviour.
- **Document Store + LLM** : To provide intelligent troubleshooting using manuals and past logs.
- **Dashboard & Alerts** : For live monitoring and early warning notifications

WHERE **LEGACY** MONITORING SYSTEM **FAILS**



Reactive Maintenance

machines are repaired only after failure occurs

Manual Inspections

technicians check equipment at fixed intervals



Threshold Monitoring Systems

trigger alerts only when values cross limits

Manual Inspections

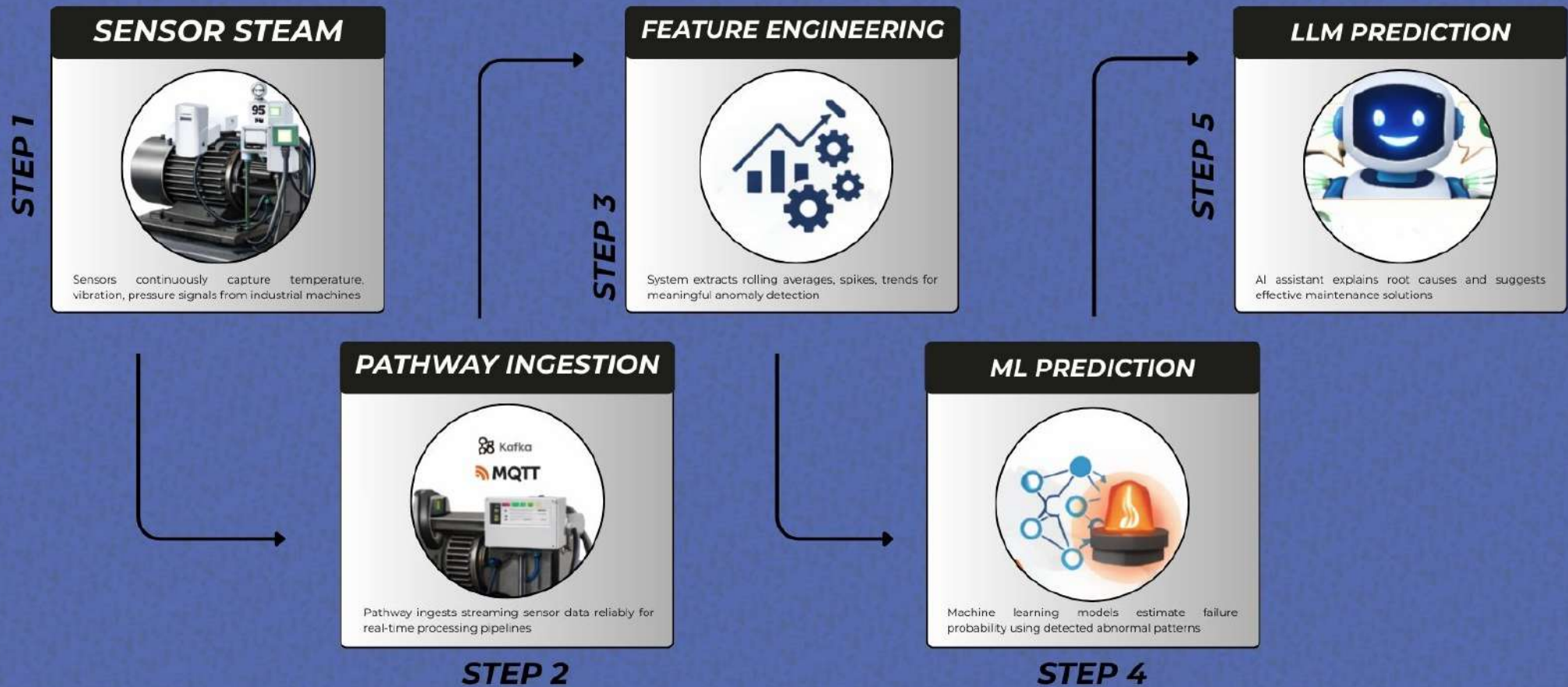
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HOW PULSE PREDICT **OVERCOMES THIS**

- Continuous **real** time monitoring instead of periodic checking.
- Predictive analytics to **identify failures** before breakdown.
- Intelligent anomaly **detection** beyond simple threshold alerts.
- **AI assistant** provides root-cause explanation and maintenance guidance.

ARCHITECTURE OVERVIEW



IMPACT AND SUSTAINABILITY

- Reduces unnecessary energy consumption caused by inefficient and faulty machines.
- Prevents catastrophic failures, minimizing material waste and environmental damage.
- Extends machine lifespan through proactive maintenance and optimized performance.
- Lowers maintenance costs by avoiding emergency repairs and production downtime.

SCALABILITY -

- Integration with real IoT sensors for deployment in live industrial environments.
- Digital twin creation for real-time virtual monitoring and simulation of equipment.
- Predictive maintenance scheduling across multiple factories and production units.
- Autonomous maintenance agents enabling self-diagnosing and self-optimizing industrial systems

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THANKYOU